

# AUGMENTED CHESS

A hands-on, AI-assisted engineering interview. Keep this open while you work.

## 1 The hour

Intro  
5 min

Build: three stages, easiest first  
about 40 min

Debrief  
15 min

The stages get harder as you go. Most engineers finish all three in the time. If you don't, that's fine: we'd rather see one stage done well than three rushed.

## 2 Ground rules

- **Use AI. Really.**  
Any assistant, agent, or editor you like. We want to see how you actually work with it.
- **Your interviewer is the product owner.**  
Ask about requirements, priorities, and edge cases whenever you like.
- **Show it works.**  
"It compiles" and "the AI says it's done" aren't evidence. Show us something that is.
- **Think out loud.**  
Tell us what you're trying, what you expect, and when you disagree with your tools.
- **You own what you ship.**  
Be ready to explain any line you commit, whoever or whatever typed it.
- **Commit after each stage.**  
`git commit -am "Stage 1"` and so on, so we can review each diff with you.

## 3 Setup

1. `git clone https://github.com/PrecisionReporters/ai_interview_challenges.git`
2. Open `augmented-chess/augmented-chess.html` in a current Chrome, Edge or Firefox. There's nothing to install, no build step and no server.
3. **Practice on one screen** plays both sides in one tab. It's the fastest way to try things out.
4. **Online play** is peer-to-peer and works between two tabs on one machine. In tab A, *Host a game* and copy the invite. In tab B, *Join a game*, paste the invite and copy the reply. Back in tab A, paste the reply and click *Connect*.

## 4 The codebase at a glance

Augmented Chess is chess where each player drafts one **augment** before the first move. There are three tiers: **Silver** bends a rule, **Gold** breaks one, and **Prismatic** rewrites the board. Some augments are **Secret** and stay hidden from your opponent until they trigger.

### <style>

All of the styling.

### Engine <script>

Rules, move generation, match lifecycle, and every augment definition.

### UI <script>

Rendering, input, sound, and the WebRTC networking between the two players.

It's one HTML file, about 2,200 lines. It works, it has some tech debt, and it has no tests. Reading it with your AI is part of the job.

**STAGE 1** WARM-UPAC-101 Feature Small

Today the game only declares a draw for insufficient material when both players are down to a bare king. Players want the standard chess rule. The game should also end in a draw for:

- king and bishop vs. king
- king and knight vs. king
- king and bishop vs. king and bishop, with both bishops on the same color

**STAGE 2** FEATURE WORKAC-102 Feature Medium

Design has handed over a new Gold augment:

**POWDER KEG**GoldPassiveSecret

"One of your pawns is secretly a powder keg. When an enemy piece captures it, the capturing piece is destroyed, along with every other enemy piece next to that square. Kings are immune. Revealed when it triggers."

Add it to the game so it can be drafted and played like any other augment.

**STAGE 3** INVESTIGATIONAC-103 Bug report High priority

From a playtester:

*"I've hosted about a dozen online games against my brother, and he always seems to know my secret augment. He walked around my landmine every single time, and he went straight past my Iron Pawn to take the one next to it. He swears he isn't cheating. Is the game broken?"*

Investigate. If something is wrong, fix it and show us it's fixed. Then tell us what you'd do next.

**What "done" means, for every stage:** the change works, you've shown us evidence that it works, and you can explain the decisions and trade-offs you made. If a requirement is unclear, ask. If you decide something yourself, say so.

**In the debrief**

We'll look at your commits together. Expect questions like: *why this approach?*, *what did your AI get wrong?*, and *what would you do with another hour?* There are no trick questions.